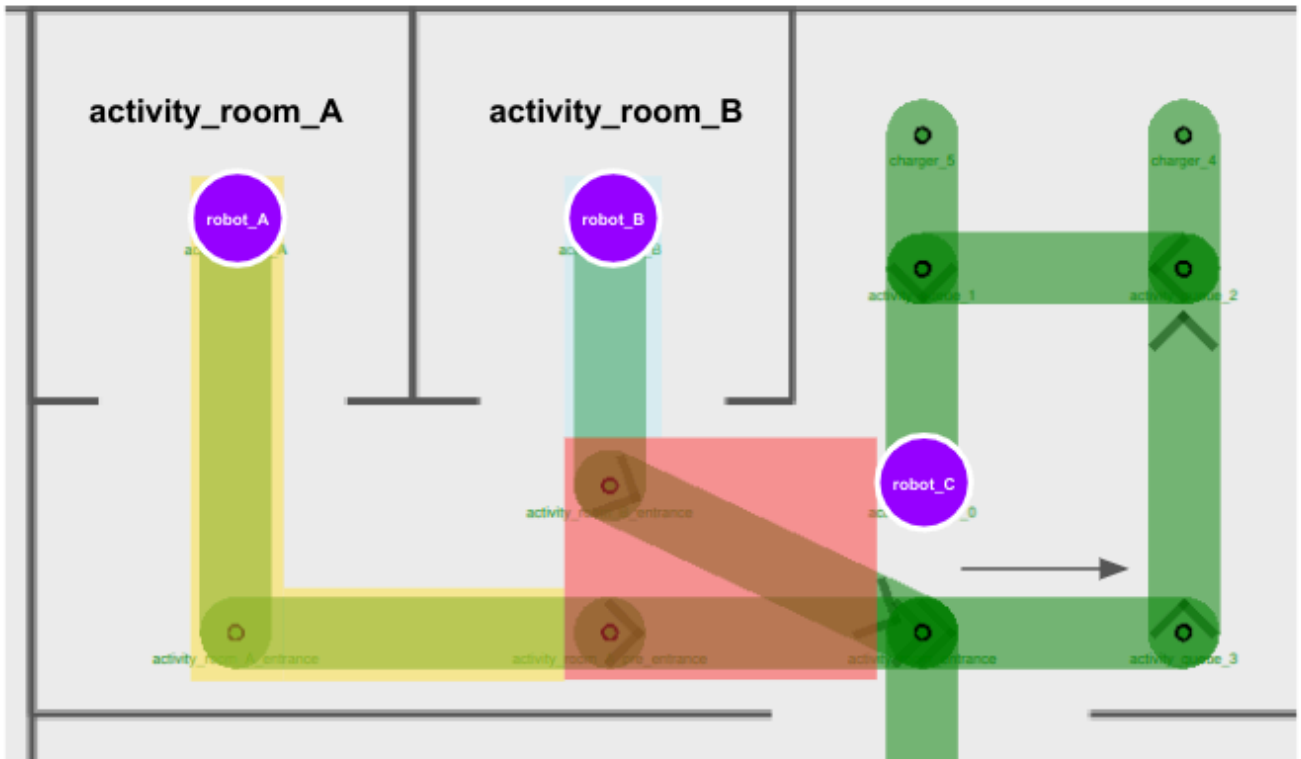
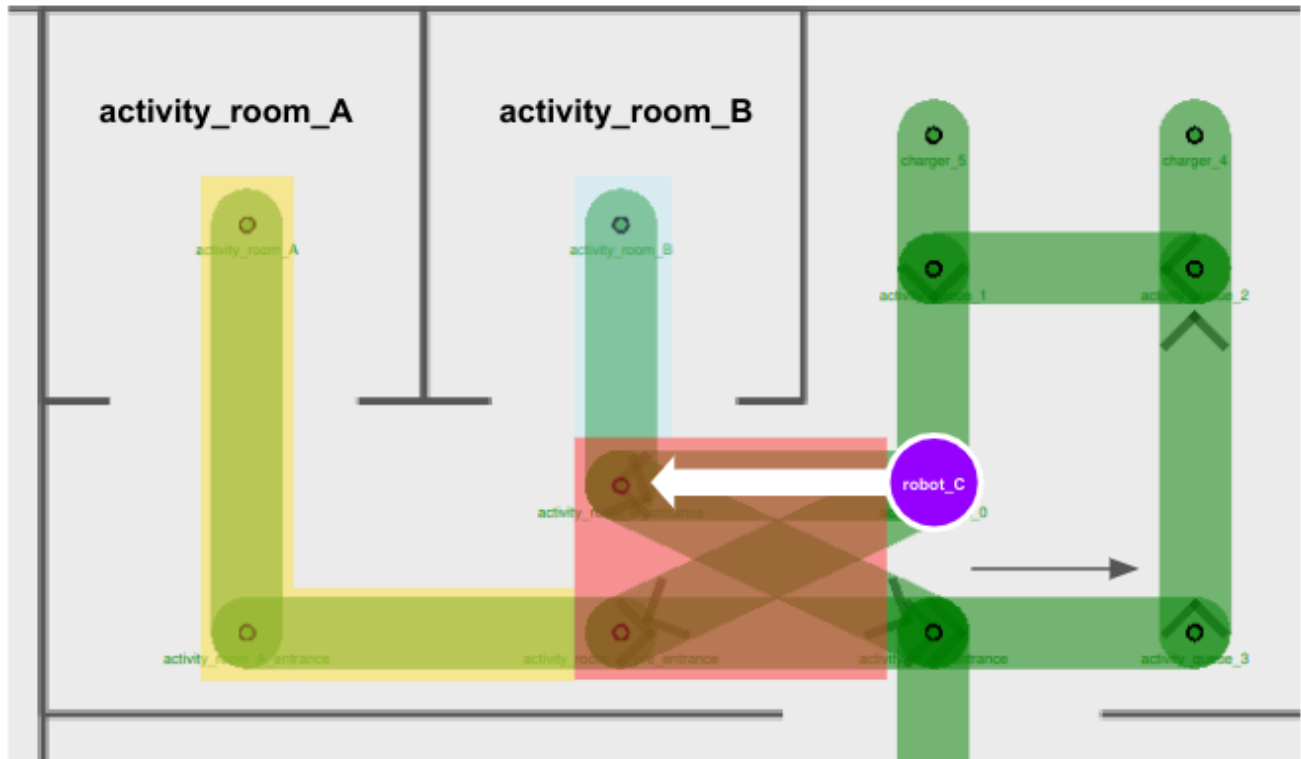


- Here is where lane closures could help. We can close off cross-section entry lanes when the respective activity rooms are occupied. `robot_C` would not be able to obtain the red mutex nor proceed with its task until `robot_B` leaves `activity_room_B`.



- After `robot_B` has successfully exited the activity room and this area altogether, the entry lane to `activity_room_B` can be opened up again for use.



Other Considerations

Instead of implementing lane closure, why don't we assign `activity_room_B_mutex` (in blue) to the entry lane towards `activity_room_B`, and `activity_room_A_mutex` (in yellow) to the entry lane towards `activity_room_A`?

We want the four cross-section lanes to belong to the same mutex group, in order to ensure that only one robot travels across any of these lanes at any point of time. This prevents situations where multiple robots move near or towards each other in a cramped corridor. Since each lane can only be assigned one mutex group, we can leverage on other features, such as lane closures, to ensure that queueing robots don't enter the cross-section if their target room is occupied.

Found a bug? [Edit this page on GitHub.](#)